

Referee Report: “Road Infrastructure and Retail Markets”

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1 Summary

This paper studies the impact of road maintenance tax cuts on road quality and housing prices. The authors compare sales prices between similar areas that narrowly pass vs. narrowly fail road property tax levy renewals in Ohio, using a sample of referendums from 1991-2021. To estimate treatment effects, they use a regression discontinuity design. They find that when local jurisdictions decide to cut road taxes, they experience an average loss of \$163,547 in road maintenance funds (11%), and this leads to a 15 percent decline in road quality after 4 years and a 9 percent drop in housing prices, relative to similar areas that opted to renew funding for road maintenance. They also find stronger effects in urban areas, as opposed to rural areas, and for more expensive houses relative to cheaper houses.

2 Main Comments

Overall Reaction: A large body of work studies the effects of transport infrastructure on a variety of different outcomes, including employment, population movements, firm locations, product prices, and innovation. Many studies have used panel data to examine how newly created surface links that expand transportation networks impact local economic activity, though some papers have also focused on quality improvements to existing roads. This paper’s attempts to link tax cuts for road maintenance to actual road maintenance expenditures, road quality, and housing prices is interesting. I also appreciate the heterogeneity analysis. At the same time, there are several aspects of the paper that I have some questions and concerns about, and others that make me uncertain about the prospects of this paper for a journal of this caliber.

1 Main Comments:

1 While I found the measurement of road quality from satellite imagery to be impressive, the discrete nature of the outcome variable seems quite coarse and tricky to interpret. What does it mean, in practice, to go from a high quality road (score = 2) to a decent quality road (score = 1), or from a decent quality road to a poor quality road (score = 0)?

1.1 Is it possible to get a sense of the mapping between quality scores and something more continuous, like road roughness? It’s hard to get a sense of this from the comparison in Figure 1. Linking your measures to road roughness would be really great, as you could

- estimate the impact of road tax cuts on travel times and speeds (conditional on speed limits), using the estimates from Currier, Glaeser, and Kriendler (2023, NBER WP).
- 1.2 The housing price estimates seem quite large, particularly compared to estimates from Currier, Glaeser, and Kriendler (2023, NBER WP) on willingness to pay to avoid road roughness. Those authors find that a one SD increase in roughness increases user costs by \$0.33 per mile, but you find that failing to renew road taxes leads to a 9% drop in housing prices. I realize both objects are different (you estimate the housing price effects of tax cuts that increase roughness, while they estimate a WTP based on slowing down and increasing driving times), but comparing these objects would be really helpful.
 - 1.3 At the very least, the change in road quality regression in Table 3 should better take the discrete nature of the outcome into account. Are there count-data analogs of the RD regressions you run? If not, can you at least do linear probability models of transitions from high (score = 2) to decent (score = 1), or from decent (score = 1) to poor (score = 0)?
 - 1.4 When jurisdictions experience losses in road maintenance funds, they have to prioritize upgrading certain types of roads more than others. It seems like your data could better shed light on the types of neighborhoods, roads, etc. within jurisdictions that are affected by the road tax cuts when those cuts are experienced. Are secondary and tertiary roads hit harder? Roads at slower speed limits? What about the neighborhoods that are affected, in terms of income and racial composition? Telling a richer story with your RD design seems possible and would be much more valuable than the average treatment effects you mostly focus on.
- 2 The housing price outcome is also a bit tricky to interpret. You are using housing price medians, but the mix of housing structures available on the market could be quite different in different municipalities at the time of the tax referendum. If this differs across the discontinuity threshold, it could bias your estimates. I'm also worried about potential confounders.
 - 2.1 Do you observe the characteristics of the houses for sale in your core logic data? If not, can you link them to other datasets (e.g. Redfin or MLS data) with that kind of information? In an ideal case, you would be able to first use a hedonic regression to strip away the impact of observable housing characteristics from the housing prices you use. Then you could take medians or quantiles in your baseline analysis.
 - 2.2 Why are you running the housing price regressions in levels? Do you get similarly sized effects with logs?
 - 2.3 Can you link within-jurisdiction variation in road quality (from my comment 1.1.4) to within jurisdiction variation in housing prices? If your local public finance theories are correct, I think we'd expect to see some dose-response relationships here.
 - 2.4 Can you look for different effects based on previous road conditions? I would expect larger effects on housing prices from moving from decent (score = 1) to poor (score = 0) than from high quality to decent.
 - 3 My other broad concerns relate to identification. Your effects seem to be quite large, though

I'm not sure how large they are relative to the rest of the capitalization literature. These large effect sizes make me concerned that you are missing important covariates, or that there are pre-trends.

- 3.1 In particular, the period you study, 1991-2021, encompasses a wide range of changes in local economic conditions, including the dot-com boom, the China trade shock, and the Great Recession. These changes presumably impacted different jurisdictions in Ohio very differently, as others have studied, and I'm not sure that a simple year fixed effect really captures those differences. In particular, harder hit places by the China shock or the Great Recession might have downward trends in housing prices and be places less likely to renew their road maintenance taxes.
- 3.2 Your covariate discontinuity exercise in Table B.1 is really helpful, but when are these variables measured? At baseline? Moreover, you are showing the lack of average cross-sectional differences across the discontinuity threshold. What about pre-trend differences? Can you look at changes in unemployment 2-3 years before, changes in population, changes in racial composition, etc.? Do these vary systematically across the threshold?
- 3.3 It wasn't clear exactly why the renewal dates are random, but new levies are not.
- 3.4 If a tax levy renewal fails, what happens? Does it disappear forever? Are there observations in the sample that have repeated failures? If so, how do you treat those observations?
- 3.5 When road maintenance tax levy renewals fail, what else happens to the budget? Does school quality also decline? What about other public services? Could your effects be picking up the impact of general tax reductions?
- 3.6 Why is there no dose-response in the effect of the tax magnitude (Figure 6)? Is there some theory that explains those results?

3 Smaller Comments

- S.1 I feel like your paper is somewhat related to Asher and Novosad (2020), but I'm not sure it's the best and most central reference. It seems like it's more closely related to the public finance / capitalization literature.
- S.2 You should relabel your appendix tables (e.g. with a letter, such as A.1, A.2) instead of continuing the numbering in the document.
- S.3 You should defer the discussion of the online retail shopping data (Section 3.3) until you Section 6 where you present the results.
- S.4 I found section 3.2 to be somewhat repetitive, as it rehashes ideas discussed earlier. I would suggest shortening or perhaps even eliminating entirely.